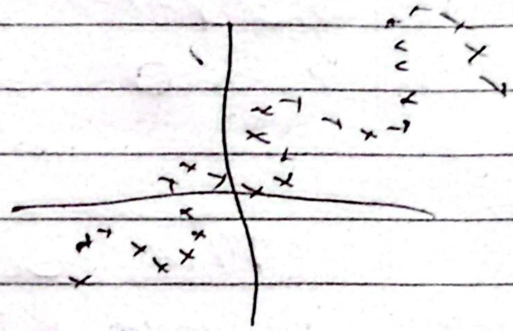
Gradient Boosting Regression.

Additive Modelling



$y = x + \sin x$

Trainingdata: $\{(x_i, y_i)\}_{i=1}^{n}$

Differentiable loss function: $L(y, F(x))$

$F(x) \rightarrow \hat{y}$

number of iterations: M.

~~Step:~~ $y = f(x)$

In ensembling:

$$f(x) = f_0(x) + f_1(x) + f_2(x) + \ldots + f_n(x)$$

→ Decision Tree.

Step 1: Initialize $f_0(x) = \arg\min_{\gamma} \sum_{i=1}^{N} L(y_i, \gamma)$

Let loss f^n: $L = \frac{1}{2} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$

Here; γ represent $\hat{y}_i$

So; $f_0(x) = \arg\min_{\gamma} \frac{1}{2} \sum_{i=1}^{N} (y_i - \gamma)^2$

$$\frac{\partial f_0(x)}{\partial \gamma} = \frac{\partial}{\partial \gamma}\left(\frac{1}{2}\sum_{i=1}^{n}(y_i - \gamma)^2\right)$$

$$\Rightarrow 0 = \sum_{i=1}^{N}(\gamma - y_i)$$

Here solving for γ gives the mean of output $[y_i]$

output of $f_0(x)$ will be mean.

Step2: for $m = 1$ to $\underbrace{M}_{\text{no. of models}}$.

$f_1(x), \ldots f_{50}(x)$ if $M = 50$

a) For $i = 1, 2, \ldots, N$ compute.

$$\underbrace{\gamma_{im}}_{\text{pseudo residual}} = -\left[\frac{\partial L(y_i, f(x_i))}{\partial (f(x_i))}\right]_{f = f_{m-1}}$$

[error distance].

y_i = ~~pred~~ output

$f(x_i)$ = predictive output.

$$\gamma_{im} = -\left[\frac{\partial}{\partial \hat{y}_i}\left(\frac{1}{2}(y - \hat{y}_i)^2\right)\right]_{\underbrace{f = f_{m-1}}_{\text{Previous answer.}}}$$

i = ~~..~~ row

m = model no.

$$\gamma_{im} = (y_i - f(x_i))_{f = f_{m-1}}.$$